MIL-P-52099B
26 August 1971
SUPERSEDING
MIL-P-52099A
30 April 1964

MILITARY SPECIFICATION

POSTS, HANDRAIL AND APPROACH

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers handrail and approach posts used with fixed or floating bridges having aluminum deck-balk superstructures.

1.2 Classification. The posts shall be of the following types, as specified:

Type I - Handrail.
Type II - Approach.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.

Military

MIL-T-704	- Treatment and Painting of Materiel.

STANDARDS

Federal

FED. STD. No. 595	- Colors.

FSC 5420

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Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

DRAWINGS

ME

DL13211E7572	- Post, Handrail.
C13211E7549	- Post, Approach.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN WELDING SOCIETY

A5.10	- Aluminum and Aluminum-Alloy Welding Rods and Bare Electrodes.
B3.0	- Standard Qualification Procedure.

(Application for copies should be addressed to the American Welding Society, 345 East 47th Street, New York, N. Y. 10017.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, N. Y. 10017.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 202, Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

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NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street, NW, Washington, D. C. 20036.)

3. REQUIREMENTS

3.1 Description. The posts shall be as shown on the data list or drawing, as applicable, and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are engineering design drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the supplier or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The supplier shall furnish one or more posts for examination and testing within the time frame specified, to prove prior to starting production that his production methods will produce posts that comply with the requirements of this specification. Examination shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.4 Identification marking. The post shall be metal-die stamped as shown on the applicable drawing. Die stamping shall be legible after painting.

3.5 Treatment and painting. The post shall be cleaned, treated, and painted in accordance with MIL-T-704, type B except that the color number designation shall be 34087 in accordance with FED. STD. No. 595.

3.6 Type I, handrail. The type I handrail post shall be as shown on DL13211E7572.

3.6.1 Welding electrodes. Welding electrodes shall conform to AWS A5.10, classification ER4043, size as required.

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3.7 Type II, approach. The type II approach post shall be as shown on Drawing C13211E7549.

3.8 Workmanship.

3.8.1 Welders and welding.

3.8.1.1 Welders. Before assigning any welder to manual welding work covered by this specification, the supplier shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operation to be performed and that such qualification is effective as defined by the particular code:

Standard Qualification Procedures of the American Welding Society (B3.0).
Welding Qualifications of the ASME Boiler and Pressure Vessel Code (Section IX).

3.8.1.2 Welding. The surfaces of parts to be welded shall be free from scale, paint, grease, and other foreign matter. Aluminum welding shall be done by the shielded metal-arc or gas metal-arc (MIG) procedures in accordance with Part B, Section IX of the ASME Boiler and Pressure Vessel Code. All welded parts shall be free from cracks and other imperfections that may reduce the effectiveness of the part. Work shall be positioned for flat welding when practicable.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspection. Inspection shall be classified as follows:

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(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of preparation for delivery (see 4.6).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction post shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the major and minor defects specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.5 Inspection procedure.

4.5.1 Examination. The post shall be examined for the following defects:

Major

101. Nonconformance to the dimensions shown on the drawings.
102. Material not as specified.
103. Workmanship not as specified.

Minor

201. Identification marking incorrect, missing, or illegible.
202. Treatment and painting not as specified.

4.6 Inspection of preparation for delivery.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

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4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

104.	Boxes not in accordance with the referenced document for level A or B.
105.	Boxes not reinforced as specified for level A or B.
106.	Strapping not zinc coated for level A.
107.	Gross weight of each box exceeds 200 pounds for level A or B.
108.	Marking illegible, incomplete, or incorrect.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or C, as specified.

5.1.1 Level A. The posts of like description shall be packed in quantities not to exceed 200 pounds gross weight. The sliding lug on type I posts shall be secured with the thumb-screw. The container shall be a close-fitting box conforming to PPP-B-621, class 2, style 2, for type 3 load. The ends of the box shall be reinforced by the addition of boards covering the entire inner face of each end. The reinforcing boards shall be the same thickness as the end boards, shall be placed with the grain direction at right angles to that of the end boards, and shall be nailed to them with the nails clinched. The sides, top, and bottom of the box shall be nailed to the reinforcing boards in addition to the nailing required by the box specification. The boxes shall be closed and strapped in accordance with the appendix to the box specification. Strapping shall be zinc coated.

5.1.2 Level B. The posts shall be packed as specified in 5.1.1 for level A except the box shall be class 1 and strapping shall not be required to be zinc coated.

5.1.3 Level C. The posts shall be packed to assure carrier acceptance and safe delivery to destination at lowest ratings in a wooden or fiberboard container complying with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.2 Marking. Shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Type I posts are to be used as support for the handrail rope and as markers to define the roadway width of aluminum deck-balk superstructure bridges. Type II posts are used to mark the bridge approach roadways for aluminum deck-balk superstructure or other types of bridges.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of post required (see 1.2).
(c) Time frame required for submission of the preproduction model (see 3.2).
(d) Level of packing required (see 5.1).

6.3 Preproduction model. Any changes or deviations of production posts from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish posts conforming to this specification.

Custodians:

Army - ME
Navy - MC

Preparing activity:

Army - ME

Project No. 5420-0118

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION **MIL-P-52099B, Posts, Handrail and Approach**

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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DEPARTMENT OF THE ARMY
U. A. ARMY MOBILITY EQUIPMENT COMMAND
RESEARCH, DEVELOPMENT AND ENGINEERING DIRECTORATE
ATTN: AMSME-RZG
FORT BELVOIR, VIRGINIA 22060

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RESEARCH, DEVELOPMENT AND ENGINEERING DIRECTORATE
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